



Case Study: Cost of a Data Breach

Estimated time: 5 minutes

Introduction

In today's digital evolution, where data is paramount, data breaches have become an unfortunate reality for enterprises of all sizes. In this reading, you'll examine the multifaceted costs associated with data breaches in an enterprise context and analyze several high-profile incidents to derive actionable insights for organizations seeking to protect their digital assets. The financial implications of data breaches extend far beyond the monetary losses, affecting an organization's reputation, operations, and long-term competitive positions.

Learning objectives

After completing this reading, you'll be able to:

- Describe the costs associated with data breaches in the organization
- Explain the financial implications of data breaches beyond monetary losses and affecting the organization's reputation
- List potential effects of data breach in the organization
- State some key considerations and recommendations to avoid data breaches in the organization

Let's understand the costs of the data breaches in GoRoam, a travel company.

Case study

GoRoam is an online ticket and hotel booking application that serves thousands of customers every day. This application is popular among students and backpackers, and customers are happy with their quick responses and the services. GoRoam has implemented a robust data architecture to protect customers' sensitive information; however, recently, GoRoam experienced a

massive data breach where millions of user profiles with their personal details, including age, email ID, phone numbers, and financial data such as credit and debit card numbers were compromised. This data breach has not only resulted in financial loss for GoRoam but also affected their reputation. Let's dive deeper to understand what it actually costs.

Costs of a data breach

The cost of a data breach in an organization is comprehensive and far-reaching, encompassing critical cost categories. These categories include a broad range of short-term and long-term expenses that can affect multiple facets of an enterprise.

- **Direct financial loss:**

Organizations face immediate financial consequences through the theft of funds, intellectual property, or valuable customer data. This situation can become more vulnerable if credit card information or other useful data is compromised. Such losses are also compounded by regulatory fines and penalties for non-compliance with data protection laws such as General Data Protection Rights (GDPR).

- **Reputation damage:**

The most devastating long-term consequence is customer trust and brand reputation erosion. Once customers feel that their personal information is compromised, they shift their business elsewhere, creating significant challenges for organizations attempting to rebuild customer loyalty and trust.

- **Legal loss:**

The after-effects of data breaches typically involve lawsuits from multiple stakeholders, partners, and shareholders. Legal fees accumulate as companies defend themselves or settle claims, while regulatory investigations add another layer of legal complexity and potential financial penalties.

- **Operation disruption:**

Significant operational downtime helps organizations to remediate the breach. The information technology (IT) team spends extensive time and resources investigating the breach, restoring systems, and patching vulnerabilities. These disruptions lead to lost productivity, delays in service delivery, and an overall negative impact on business operations.

- **Customer compensation and support:**

Organizations should compensate the affected customers in various forms, including credit monitoring services, identity theft protection, or direct monetary

settlements. Additional costs may arise from establishing dedicated customer support teams to manage inquiries and assist with recovery efforts.

- **Loss of competitive advantage:**

In certain cases, where intellectual property or proprietary business information is compromised, organizations may lose their market edge. Competitors may use the stolen information to develop similar offerings, undermining the breached company's position in the marketplace.

- **Increased cybersecurity investments:**

Organizations typically make substantial investments in enhanced security measures to prevent future thefts. They invest in new cybersecurity software, infrastructure upgrades, and additional cybersecurity personnel. These investments are crucial but add financial burden already created by the breach.

- **Higher insurance premiums:**

Organizations with cyber insurance often face high premium rates after the data breach as their insurers reassess their risk profile and adjust coverage costs accordingly.

Now, let's review certain case studies with high data breaches.

Examples of high-profile data breach

Several high-profile data breaches have occurred over the years, and some of them have resulted in enormous financial losses.

1. Yahoo (Year: 2013–2014)

- **Cost impact:** \$350 million
- **Scope:** 3 billion user accounts compromised
- **Key implications:** A \$350 million reduction in the purchase price during Verizon's acquisition of Yahoo highlights immense costs associated with a breach of this magnitude
- **Primary failure:** Delay disclosure and underestimating breach sensitivity

2. Equifax: Year 2017

- **Cost impact:** \$700 million
- **Scope:** Personal data of 146 million individuals exposed, including social security numbers (SSNs) and credit card details

- **Key implications:** Faced numerous lawsuits and penalties, leading to the estimated total cost of \$700 million for settlements, fines, and other expenses
- **Primary failure:** Unpatched software exposures

3. Target: Year 2013

- **Cost impact:** \$162 million
- **Scope:** 40 million credit and debit card customers are affected
- **Key implications:** Paid millions in legal fees, settlements, and fines, damaged brand image
- **Primary failure:** Compromised vendor credentials and insufficient network segmentation

4. Home Depot: Year 2014

- **Cost impact:** \$179 million
- **Scope:** Compromised details of 56 million payment cards
- **Key implications:** Paid millions in legal costs, customer notifications, and boosting cybersecurity, lost customers' trust
- **Primary failure:** Compromised customer sensitive information

5. Anthem: Year 2015

- **Cost impact:** \$115 million
- **Scope:** Compromised personal information for 80 million patients
- **Key implications:** Paid millions as settlement costs, compensation costs to patients, and investment in cybersecurity
- **Primary failure:** Not invested in cybersecurity measures

6. Marriott Internationals: Year 2018

- **Cost impact:** \$115 million
- **Scope:** Personal details of 500 million customers compromised
- **Key implications:** Paid significant legal fees, customer support costs, and regulatory fines
- **Primary failure:** Inadequate due diligence during the acquisition of Starwood, where the breach originated

7. Facebook: Year 2019

- **Cost impact:** \$5 million
- **Scope:** Privacy violation affecting millions of users by Cambridge Analytica Scandal, where the personal information of millions of users was misused
- **Key implications:** Record-breaking fine of \$5 million charged by Federal Trade Commission (FTC)
- **Primary failure:** Insufficient data governance and third-party access controls

8. Sony Play Station Network: Year 2011

- **Cost impact:** \$171 million
- **Scope:** 77 million users personal information compromised
- **Key implications:** Paid millions in legal fees and customer compensation and invested in rebuilding online infrastructure
- **Primary failure:** Lack of poor online infrastructure

9. Uber: Year 2016

- **Cost impact:** \$148 million
- **Scope:** 57 million users' and drivers' personal information compromised
- **Key implications:** Paid millions in significant legal challenges and settlements for the affected users
- **Primary failure:** Lack of poor cybersecurity measures

10. Capital One: Year 2019

- **Cost impact:** \$80 million
- **Scope:** 100 million customers personal information compromised in the United States and Canada
- **Key implications:** Paid millions in legal fees, customer notifications, and invested in cybersecurity enhancements
- **Primary failure:** Misconfigured firewall, highlighting risks posed by simple security oversights

Key lessons and recommendations

Based on the evaluation of the high-profile data breaches and their associated legal costs, organizations should consider the following protective measures:

- Implement comprehensive security governance by establishing clear policies, regular audits, and accountability structures
- Adopt a defense-in-depth strategy to protect critical assets, ensuring no scope of failure
- Maintain robust patch management systems by promptly addressing software vulnerabilities to avoid common cyberattacks
- Perform regular penetration testing and vulnerability assessments to identify weaknesses before attackers identify them
- Educate employees about cybersecurity best practices and social engineering threats
- Implement a strong vendor management system by assessing third-party security practices and limiting access to sensitive information

Summary

In today's digital economy, data breaches represent a significant and multifaceted threat to organizations. The organizations face financial losses, affecting multiple aspects of business operations and long-term viability. Therefore, organizations should protect their digital assets and maintain stakeholders' trust in an increasingly vulnerable digital landscape.